QISKIT FALL FEST 2026 · HACKATHON SUBMISSION

QuakeNet-Q

An autonomous agent that uses quantum optimisation to decide where the first portable seismometers go after a major earthquake

DOMAIN	Disaster Response
TEAM	Muhammad Abdullah, Muhammad Hassaan-ul-Mustafa, Rafia Ali
QUANTUM METHOD	Dicke-state, XY-mixer QAOA in Qiskit 2.6.2, trained on CVaR, checked by exact search
DATA	Real USGS aftershock catalogues and IRIS station lists; six earthquakes
CODE	https://github.com/MuhammadAbdullah80/quakenet-q
LIVE CONSOLE	https://quakenet-q.vercel.app
DATE	12 November 2026

In one paragraph

After a large earthquake, response teams have hours to decide where a handful of portable seismometers go. QuakeNet-Q watches the USGS feed, starts itself, builds an exact score for every possible network, lets a quantum circuit choose 4 of 10 sites, checks the choice against every alternative, and re-plans hourly as aftershocks arrive. On the 2019 Ridgecrest earthquake the quantum step found the best network in 15 of 15 runs. It is a proof of concept: no quantum speedup is claimed, and the limitations in Chapter 9 are part of the result.

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Executive summary

The problem. When a major earthquake strikes, the permanent seismic network is usually too sparse to locate the thousands of aftershocks that follow. Teams therefore rush a few portable seismometers into the field within hours. Where those instruments go decides how sharply the aftershocks can be located, and sharp locations show which faults are still active, which feeds aftershock forecasts and decisions about where rescuers and residents are safe. In the published deployment reports we found, the sites are chosen by expert judgement and simple rules, under time pressure.

What we built. QuakeNet-Q is an automated program (not an AI or chatbot) that makes this decision and keeps it up to date. It reads the USGS feed, starts itself when a qualifying earthquake occurs, plans a first network within minutes, then checks every hour whether the aftershocks have moved and re-plans if they have. Every plan is made by a quantum algorithm written in Qiskit and checked against an exact search over all possible networks. A web console lets anyone test each part by hand.

Why the quantum part is real work. The choice of sites is made by the quantum circuit, not decorated by it. We encode the seismological score exactly (all 385 terms up to groups of four sites), start the circuit in a Dicke state that holds only valid networks, and use an XY mixer that never leaves them. A classical optimiser tunes the circuit; classical code then re-scores and verifies the answer.

Table 1. *Headline results. Every number is produced by a script in the repository.*

Question	Result	Source
Does the circuit encode the true score?	Yes: zero error on all 210 networks (385 terms)	run_experiment.py
Does the quantum step find the best network?	15 of 15 runs on Ridgecrest; 14 of 15 on five more earthquakes (CVaR)	run_experiment.py, run_benchmarks.py
How often does one measurement give the best network?	5.1% to 11.0% (random: 0.48%); our 20% target was **missed**	run_experiment.py
Does the chosen network locate aftershocks better?	Ridgecrest: 3.00 km with no new sensors, 2.63 km expert rule, **2.31 km** our pick (simulated test)	run_experiment.py
Does it beat the expert rule elsewhere?	**Only on 1 of 5 earthquakes**; Puerto Rico 2020 was 15.8% worse	run_benchmarks.py
Does the agent work end to end?	A 24-hour Ridgecrest replay ran in 23 s and made 3 plans	run_extras.py
What would hardware noise do?	At 1% gate error the chance of the best network fell from 52% to 24% (small simulated case)	run_extras.py

What we do not claim

No quantum speedup: with 210 possible networks a laptop checks every one in milliseconds, and greedy search also finds the optimum. The instance is kept small on purpose so that every answer can be verified.

No field validation: the location test is a simulation, and live mode has not yet run during a real qualifying earthquake.

The score we optimise does not always beat a simple field rule (Section 8.1). Better placement on those earthquakes needs a better score, not a better search.

1. The problem and who it serves

1.1 What happens after a big earthquake

A large earthquake is followed by thousands of aftershocks over days and weeks, some of them damaging. Seismologists locate each aftershock from the times its waves reach different seismometers, much as a phone can be located from the timing of signals at several towers. If all the instruments sit on one side of the aftershocks, the locations are blurred. The permanent network is rarely dense enough near a new rupture, so agencies such as the USGS deploy portable instruments within hours to days (Cochran et al. 2020).

Optimal network design has been studied since the 1970s (Kijko 1977; Rabinowitz & Steinberg 1990), and Hardt & Scherbaum (1994) designed optimum aftershock networks specifically. But the deployment accounts we read describe siting in qualitative terms: improve azimuthal coverage, get near-field observations, work around road access and permits. Tools that do exist, such as SN-CAST, the AuScope network design app and NetOpt3D, are built for planning a network in advance; we did not find one that starts by itself after an earthquake and re-plans as the aftershocks evolve.

1.2 Problem statement

> **Problem statement**
>
> Given the aftershocks recorded so far, the permanent stations, and a set of candidate sites, choose where to put a limited number of portable seismometers so that later aftershocks are located as accurately as possible, without wasting instruments on redundant sites and without sending crews to needlessly risky places. Do this automatically within minutes of the earthquake, and update the choice as the aftershocks move.

Table 2. *The problem in the terms of the hackathon brief.*

Element	In this project
Users	Rapid-response seismology teams (for example USGS, INGV, GFZ) and the emergency managers who rely on their aftershock maps
Limited resource	4 portable instruments, 10 candidate sites
Competing objectives	Location accuracy against sensor redundancy against crew risk. The most informative sites lie on the active rupture, which is also the hardest and most dangerous place to work
Decision speed	A first plan within minutes, before many real aftershocks exist; re-plans every hour
Success measure	Mean error when locating later aftershocks that the planner never saw

2. How the idea developed

The brief asks teams to choose their own problem. We did not start with seismometers. This chapter records the stages we went through, the ideas we rejected and why, and the corrections we made along the way, because the path explains several design choices that would otherwise look arbitrary.

2.1 Stages at a glance

Table 3. *The twelve stages from brief to submission.*

#	Stage	What we did	Outcome
1	Read the brief	Translated the problem statement and the six domains into plain language; learned what QUBO, QAOA, a qubit budget and an exact baseline mean	A shared vocabulary; the rule of thumb that a laptop simulates about 10 to 16 qubits, so one qubit per yes/no decision
2	First domain: UAVs and robotics	Mapped 12 sub-areas (task allocation, routing, scheduling, charging, coverage, swarms, airspace, payload, bases, relays, warehouses, inspection) and generated ideas for each	Ideas were sound but generic: drone routing and task assignment are what most teams would submit
3	Search for non-generic problems	Asked for research-led problems across all six domains and tagged each as a research gap (no quantum paper found) or research-backed (a paper to build on)	About 15 candidates, from sterile-insect release drones to proton-therapy beam angles
4	Shortlist and hybrid test	Took five favourites and asked of each: can it be hybrid, with one decision type and one simple rule, scored by a physical model, small enough to verify exactly?	Seismometer siting ranked first; intercropping dropped (too many penalty terms)
5	Gap check	Asked why no seismometer placement agent exists; searched both the seismology and the quantum literature	Classical methods since the 1990s, quantum sensor placement in other fields, no link between them
6	Deep research	Three parallel searches (classical lineage, quantum gap, data and Qiskit feasibility), about 30 sources, four hypotheses tested	Gap and data confirmed; one source we had relied on was wrong for our case (2.4); self-assessed fit with the brief 8.6 of 10
7	Judge's view	Asked how a judge with no seismology would see it	Reframed as a rescue story with a 'blur' metric and a phone-location analogy; jargon kept for the report
8	Version 1 proof of concept	Pairwise QUBO on real Ridgecrest data, XY-mixer QAOA, brute-force and greedy baselines	Worked, but two weak points: an approximate score and only model-predicted errors
9	Critique and upgrade	The team asked for an autonomous, self-checking agent rather than a support tool, and for the highest success rate possible	Exact 385-term score, independent location test, CVaR training, more earthquakes, the agent loop
10	Technique comparison	Compared our method with two published network-design systems (Strutz & Curtis 2024; NetOpt3D)	Adopted hearing range and a slope filter; rejected 3D ray tracing and genetic algorithms at our size
11	System decisions	Chose how the agent is hosted and tested	FastAPI on the team laptop, a Next.js console on Vercel, no AI or LLM features anywhere
12	Build and measure version 2	Built and tested the agent, ran five more earthquakes and a noise study	The results in Chapters 7 and 8, including the ones that did not go our way

2.2 The ideas we considered

In stage 3 we deliberately looked for problems that real specialists work on but that hackathon teams rarely pick. Table 4 lists the shortlist from stage 4 with the assessment we made at the time.

Table 4. *The five shortlisted ideas (stage 4).*

Idea	Domain	Unique	Hard to criticise	Impact	Effort	Decision
Aftershock seismometer network	Disaster Response	High	High	High	Medium	**Chosen.** One decision type, one rule, a physical score, a real dataset
Proton-therapy beam angles	Healthcare	High	Medium to high	Very high	Medium to high	Runner-up. Needs a toy patient model, which invites the criticism that it is not real
PMU placement in a power grid	Energy	Low	Very high	Medium	Low to medium	Safest, but well studied; the coverage rule needs penalty terms
In-pipe robot entry points	Robotics	High	Medium	Medium	Medium	Pairwise overlap is only an approximation of true pipe coverage
Intercropping layout	Agriculture	Medium	Low	Medium	Medium	Dropped: 'one crop per plot' needs many penalties that are easy to get wrong

Ideas considered earlier and set aside included sterile-insect release drones, mass-casualty triage to hospitals, kidney paired exchange, beehive placement for orchards, cold-chain vaccine carriers, drones as emergency cell towers and urban drone altitude lanes. Several were promising but either had published quantum solutions already, or needed penalty-heavy formulations that a judge could easily pick apart.

Why seismometer siting won

- **It fits the quantum method naturally.** 'Choose exactly k of n sites' is one decision type with one constraint, which a Dicke state and XY mixer enforce without any penalty weight to tune.
- **The score is physics, not invented numbers.** The D-criterion comes from the seismology literature and can be computed from public data.
- **It can be judged independently.** Locating later aftershocks gives an error in kilometres that is separate from the score we optimise.
- **Real data is free and keyless.** USGS and IRIS returned the Ridgecrest catalogue and station list on the first try.
- **It is fresh.** We found no published quantum formulation of seismic station placement (Section 2.3).

2.3 What the research established

Table 5. *Hypotheses tested before building (stage 6).*

Hypothesis	Verdict	Basis
No published quantum (QUBO, QAOA or annealing) formulation of seismic station placement exists	Not refuted	About eight targeted searches found none. The closest work is quantum seismic inversion (Nguyen & Tura 2024) and quantum sensor placement on bridges, water networks and factories. Google Scholar and non-English literature were not searched, so we say 'first we found', never 'first ever'
Rapid aftershock deployments choose sites by judgement and logistics, not an optimiser	Leaning confirmed	USGS Ridgecrest (Cochran et al. 2020), Italy's SISMIKO, Turkey 2023 and EarthScope RAMP describe qualitative goals only. Full texts were not all available
The seismological score can be written in a form a quantum circuit can optimise	Confirmed, then improved	Version 1 used a pairwise approximation; version 2 uses the exact expansion up to four-site terms

Hypothesis	Verdict	Basis
Open data is enough for a realistic instance	Confirmed	USGS ComCat and the IRIS station service, no key needed; Qiskit 2.6.2 supports every gate we need

2.4 Corrections we made along the way

Being honest about the project includes being honest about our own mistakes. These are the ones that changed the design:

- **A borrowed formulation did not fit.** We first planned to use a mutual-information QUBO (arXiv 2407.14747). On reading it closely, it is quadratic only for three sensors and becomes higher-order beyond that. We switched to the pattern of San Martín et al. (Dicke state plus XY mixer).
- **'No tool exists' was too strong.** Pre-deployment design tools (SN-CAST, AuScope, NetOpt3D, Strutz & Curtis) do exist. The claim became: no tool we found is built for aftershocks, starts itself, or re-plans.
- **An early headline number was a prediction, not a measurement.** Version 1 reported a drop in 'blur' from 2.73 km to 1.50 km. That was the model's own predicted uncertainty. Version 2 replaced it with a separate simulated location test using a different wave speed and added timing noise.
- **A confidence score could blow up.** The agent measured drift relative to the net score, which can be near zero; one error read 988% and could have triggered false re-plans. Drift is now measured relative to the predicted information gain.
- **An AI layer was proposed and withdrawn.** The team decided the app has no AI or LLM features; 'agent' means an automated program.

2.5 Technique comparison with published systems

Before version 2 we compared our method with two published network-design systems: Strutz & Curtis (2024), which uses a Bayesian information score and a genetic algorithm, and NetOpt3D (Keil et al. 2023), which uses ray tracing and signal-to-noise modelling.

Table 6. *What we took from published network-design systems (stage 10).*

Technique	Decision	Reason	Status
Signal-to-noise dependent timing error ('hearing range')	Adopt, simplified	Distant sites cannot record small aftershocks	Built, opt-in
Slope filter for candidate sites (under 20°)	Adopt	Steep sites are unusable; keyless terrain tiles	Built
Bayesian DN score next to the D-criterion	Adopt as a candidate	Better in one study, but all evidence is from one group	Computed in the experiment; not used by the agent
Nested Monte Carlo check of top designs	Adopt	Guards against DN's failure on clustered aftershocks	Not built
Genetic algorithm or annealing as the main search	Reject	Exhaustive search is exact at 210 networks	Annealing kept as a baseline
3D ray tracing	Reject	Needs 3D Earth models and heavy solvers	Stated as a limitation
Reuse of published code (WoWED)	Reject	GPL-3.0 licence would bind our whole project	Own implementation

3. Formulation

3.1 Test instance

Table 7. *The Ridgecrest test instance (real data).*

Item	Value	Source
Earthquake	M7.1 Ridgecrest, California, 6 July 2019	USGS ComCat
Planning data	550 aftershocks (M ≥ 3) in the first 12 hours	USGS ComCat
Test data	217 later aftershocks (to 13 July), never seen by the planner	USGS ComCat
Permanent stations	14 stations of network CI	IRIS / EarthScope FDSN
Candidate sites	10 grid nodes at least 6 km from any permanent station	Our rule (3.3)
Budget	k = 4 sensors, so 210 possible networks	Assumption

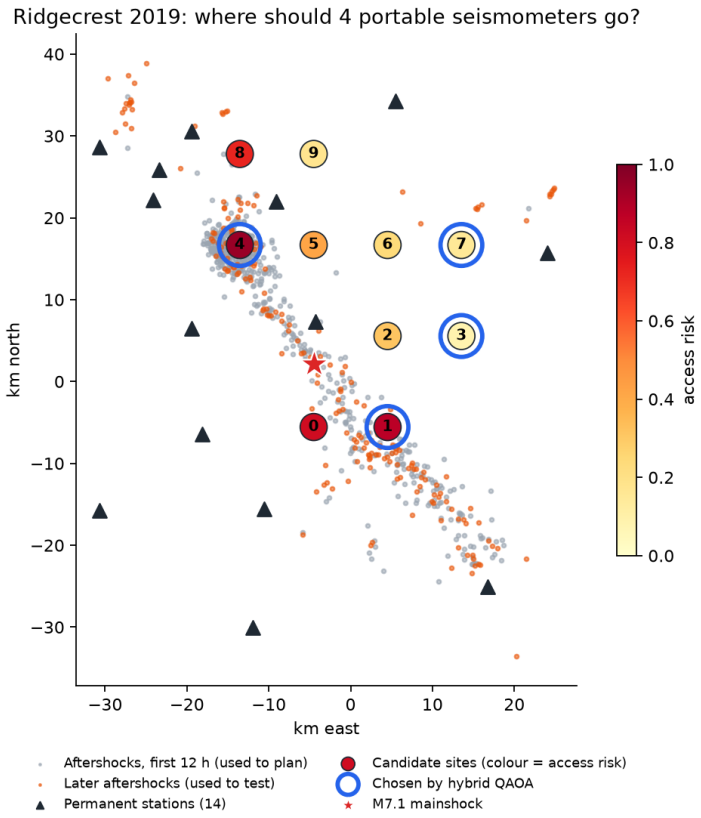


Figure 1. *The instance. Grey dots were used for planning, orange dots for testing. Numbered circles are candidate sites coloured by access risk; blue rings mark the sites the hybrid QAOA chose.*

3.2 Variables, objective and constraint

Table 8. *The optimisation problem.*

Part	Definition
Decision variables	x_i ∈ {0, 1} for i = 0…9; x_i = 1 means a sensor goes to candidate site i. One qubit per site
Constraint	Σ x_i = 4 (exactly 4 sensors)
Information score	D(S) = Σ_e log det F_e(S), the D-optimality criterion. F_e is the 4×4 Fisher information matrix of aftershock e's position and origin time, from P-wave arrival times at the permanent stations plus the chosen sites. A larger determinant means a smaller error ellipsoid

Part	Definition
Access risk	c_i = exp(−d_i / 5 km), where d_i is the distance from site i to the nearest early aftershock
Objective	maximise D(S) − λ Σ c_i x_i with λ = 150

The log-determinant is not a quadratic function, so it does not fit the usual QUBO form. Because the circuit only ever holds networks of exactly four sites, we can still encode it exactly, using its inclusion–exclusion (Möbius) expansion up to groups of four sites:

$$f(x) = \Sigma_{T,\ |T| \le 4} w_T \Pi_{i \in T} x_i - \lambda \Sigma_i c_i x_i$$

- $w_{i} = D(\{i\}) - D(\emptyset)$: the information site i adds on its own.
- $w_{i,j} = D(\{i, j\}) - D(\{i\}) - D(\{j\}) + D(\emptyset)$: how two sites interact; negative when they are redundant. Groups of three and four follow the same pattern.
- With 10 sites this gives 385 terms, and on every four-site network the sum equals the true score exactly.

Substituting $x_i = (1 - Z_i)/2$ turns f into a Hamiltonian of Z, ZZ, ZZZ and ZZZZ terms whose lowest-energy state among four-site networks is the best network.

3.3 Assumptions

- A uniform P-wave speed of 6 km/s and a picking error of 0.1 s (the agent averages three speeds: 5.5, 6.0, 6.5 km/s).
- Candidate sites come from a regular grid; nodes within 6 km of a permanent station are dropped and the 10 closest to the aftershock cloud kept. In the benchmarks, candidates also pass a road-access and slope check.
- The access-risk rule is our own proxy. A real deployment would use roads, permits and safety assessments.

4. The quantum algorithm

We use the Quantum Approximate Optimization Algorithm (QAOA; Farhi et al. 2014) in its constraint-preserving form, the Quantum Alternating Operator Ansatz (Hadfield et al. 2019). Each part has a specific job:

Table 9. *The parts of the hybrid algorithm.*

Component	What it does	Why
Dicke state $\|D(10,4)\rangle$	Starts the circuit in an equal superposition of exactly the 210 valid networks (deterministic circuit of Bärtschi & Eidenbenz 2019)	No probability is wasted on networks with the wrong number of sensors
Cost layer $\exp(-i\gamma H)$	Gives each network a phase proportional to its score; multi-qubit Z terms use a CX ladder, one RZ, and the ladder undone	Encodes the exact score, including four-site terms
XY ring mixer	$\exp(-i\beta(XX+YY)/2)$ on neighbouring qubits swaps a chosen site with an unchosen one	The number of chosen sites never changes
Classical optimiser	COBYLA tunes the 2p angles	Standard derivative-free choice for noisy objectives
CVaR training ($\alpha = 0.1$)	Tunes for the mean of the best 10% of outcomes rather than the average	Raised exact final picks from 8/15 to 14/15 in the benchmarks
Warm start	Starts from angles that worked on earlier earthquakes, kept in memory	Starts closer to a good answer

Component	What it does	Why
Sampling and check	1,024 measurements on Qiskit's StatevectorSampler; the top networks are re-scored with the true score and compared with exhaustive search	The published answer is always verified

Why not the textbook penalty approach? Adding a penalty $P(\Sigma x - 4)^2$ lets the circuit put probability on invalid answers and requires tuning P. San Martín et al. (arXiv:2305.08738) found XY mixers with Dicke states better for sensor placement on structures; we tested the same question on our problem (Section 7.2).

5. System architecture

5.1 One plan: the hybrid pipeline

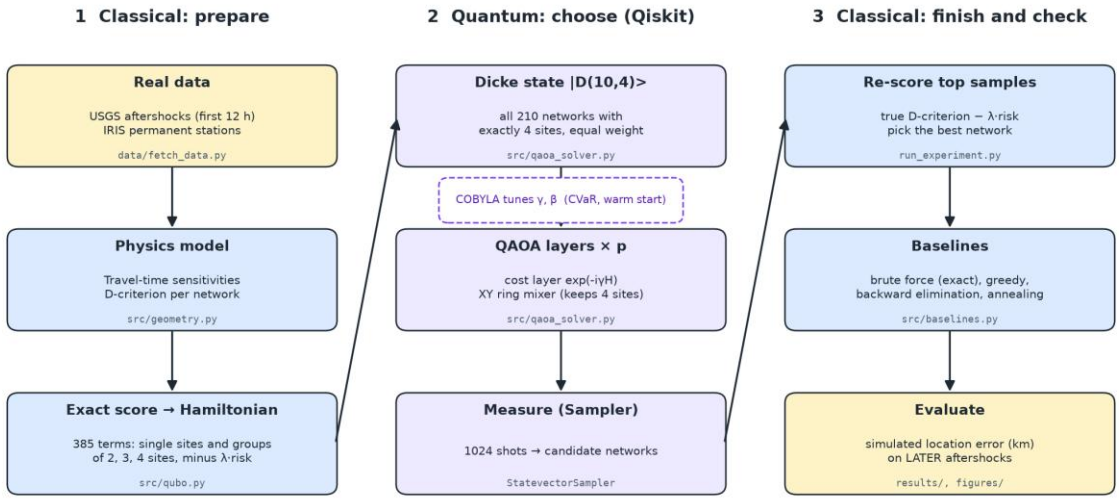


Figure 2. *How one plan is made. Classical code builds the exact Hamiltonian, the Qiskit circuit chooses the sites, and classical code re-scores and verifies the result.*

5.2 The whole system

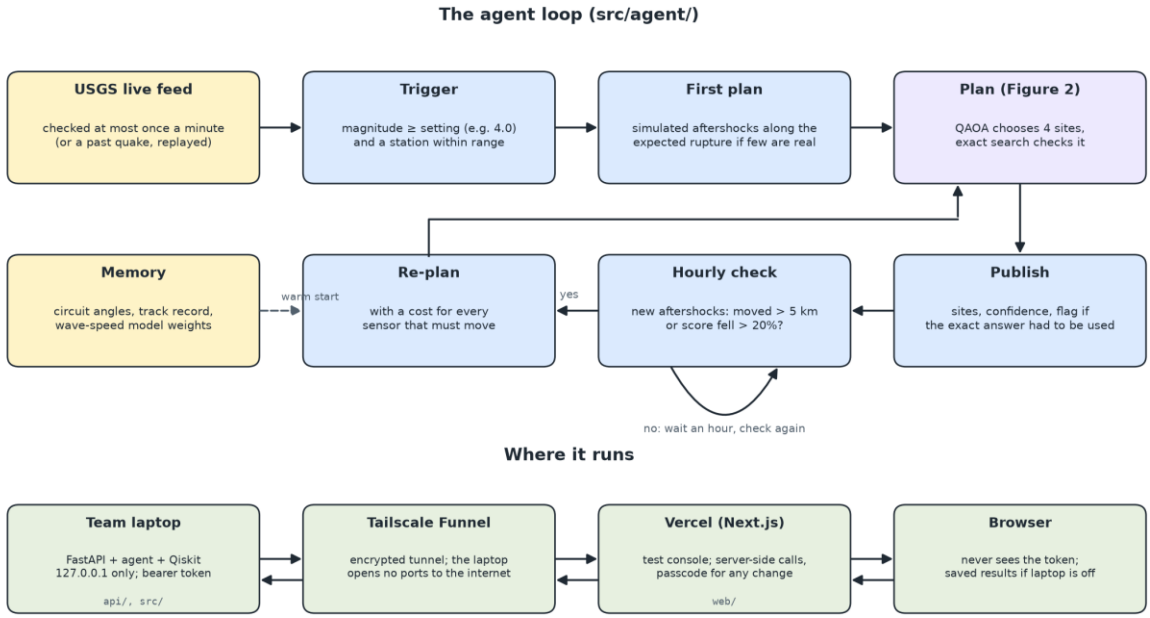


Figure 3. *The agent loop and where each part runs. The quantum step (Figure 2) sits inside the loop; the browser never talks to the laptop directly.*

Table 10. *Components.*

Part	Role	Where in the code
Data gateway	Fetches and caches USGS catalogues, station lists and terrain tiles; one place for rate limits, retries and a circuit breaker	data/, src/sources.py
Science layer	Travel times, D-criterion, hearing range, exact objective, candidate pools	src/geometry.py, src/qubo.py, src/candidates.py
Quantum solver	Dicke state, XY mixer, cost layer, CVaR training, sampling	src/qaoa_solver.py
Baselines	Exhaustive search, greedy, backward elimination, annealing, expert rule	src/baselines.py
Agent	Settings, trigger, first plan, hourly checks, re-plans, memory	src/agent/
Web API	FastAPI with a bearer token, rate limits and a job queue; bound to 127.0.0.1	api/
Test console	Next.js pages for every module; server-side calls; passcode for any change	web/
Tests	303 offline checks for the Python code (about 45 s) and 10 for the console	tests/, web/lib/

5.3 Hosting and security

- The backend runs on the team laptop and listens only on 127.0.0.1. A Tailscale Funnel tunnel exposes it over HTTPS without opening ports; every request needs a shared bearer token.
- The console on Vercel calls the backend from the server only, so the token never reaches a browser. Any change (settings, starting a run) needs a passcode, checked in an HTTP-only cookie with same-origin checks.

- Secrets live in environment variables only and are never committed. With the laptop off, the console shows saved results.

6. The autonomous agent

The experiment in Chapter 7 is one decision. The agent makes that decision automatically and keeps it up to date.

Table 11. *What the agent does after an earthquake.*

Step	What happens	Setting (default)
Trigger	An earthquake at or above the trigger magnitude, with a permanent station within range, starts a run	Trigger magnitude, editable from 3.0 to 9.0; live watching is off by default
First plan	If fewer than 30 real aftershocks exist, it simulates them along the expected rupture (length from Wells & Coppersmith 1994, magnitudes from a Gutenberg–Richter law) and plans on those	Minimum real aftershocks
Plan	The quantum circuit chooses the sites; exhaustive search checks them. After two failed retries (each with two more layers and 4,096 shots) the exact answer is published and flagged	Retries, layers, shots
Hourly check	Compares the plan with the aftershocks recorded since. Re-plans if their centre moved more than 5 km or the plan's score fell more than 20%	Interval, duration, thresholds
Re-plan	Adds a cost for every sensor that must move, so it does not shuffle crews for small gains	Move cost (40)
Memory	Keeps circuit angles that worked, the track record of predicted against realised scores (which sets a confidence level), and weights for three wave-speed models	Stored on the laptop

6.1 A 24-hour replay of Ridgecrest

A past earthquake can be played forward hour by hour with a simulated clock that never uses later data. With default settings and an empty memory, a 24-hour Ridgecrest replay took 23 seconds and made 3 plans:

Table 12. *Plans made during the replay (run_extras.py).*

Hour	Event	Aftershocks used	Sites	Why	Quantum agreed with exact
0	First plan	simulated	{3, 7, 8, 9}	no real aftershocks yet	Yes
1	Re-plan	236 real	{2, 3, 7, 9}	centroid moved 7.9 km; score fell 44%	Yes
23	Re-plan	1249 real	{1, 3, 4, 7}	centroid moved 5.1 km	Yes

The agent ended on sites {1, 3, 4, 7}, the same network the full experiment finds with 12 hours of data, after moving 3 sensors in total.

7. Results on Ridgecrest

7.1 The exact score against the usual pairwise approximation

Table 13. *How well each version of the score matches the true score.*

Terms kept	Terms	Rank correlation with true score	Largest error	Rank of its optimum	Gates per layer
Groups up to 2 sites	55	0.9873	98.7	1	908
Groups up to 3 sites	175	0.9995	17.0	1	1508
Groups up to 4 sites	385	1.0000	0.0	1	2978

Keeping groups of up to four sites reproduces the true score of all 210 networks with zero error. The pairwise version ranks networks well (ρ = 0.987) and shares the optimum at λ = 150, but at λ = 300 its optimum is only number 2. The exact version costs about 3.3 times as many gates per layer. We chose correctness.

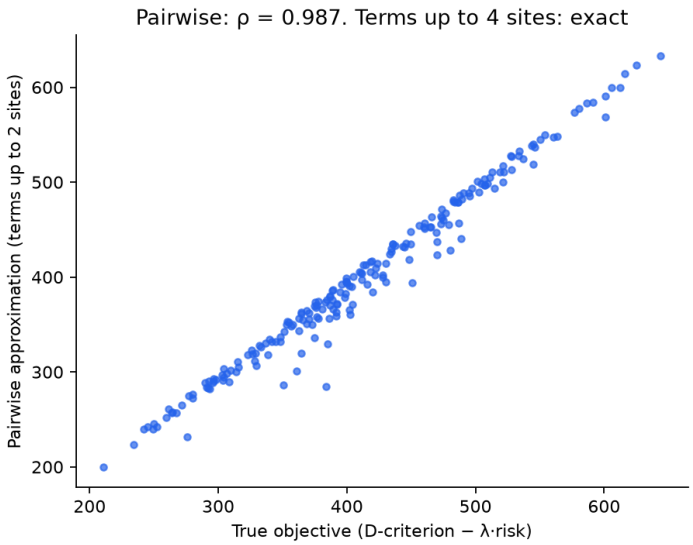


Figure 4. *Pairwise approximation against the true score for every network. The exact expansion lies on the diagonal.*

7.2 Does the quantum circuit work?

Table 14. *QAOA performance over 5 seeds per depth. The penalty circuit is the textbook alternative.*

Layers	Approx. ratio	P(best network), XY	vs random	Shots for 95%	Valid, penalty	P(best), penalty
1	0.735	5.1% (5.1%–5.1%)	11×	58	52%	0.27%
2	0.768	8.3% (6.9%–9.2%)	18×	35	68%	0.36%
3	0.788	11.0% (9.0%–12.6%)	23×	26	73%	0.37%

A random valid network is the best one with probability 1/210 = 0.48%. The XY-mixer QAOA raises that to 5.1% with one layer and 11.0% with three, and every measurement is a valid four-site network. After re-scoring, the pipeline returned the exact optimum, sites {1, 3, 4, 7}, in 15 of 15 runs. Our own target was 20% per measurement; we did not reach it.

The penalty circuit gave a valid answer only part of the time and measured the optimum less often than random guessing among valid networks. We used one untuned penalty weight (P = 2 in scaled units), so a better-tuned penalty circuit could do better than ours did.

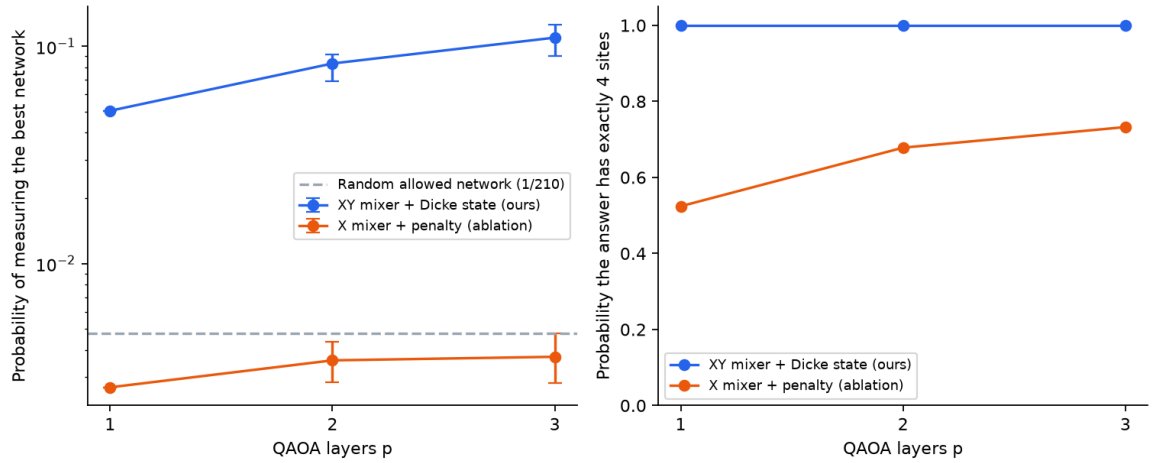


Figure 5. *Left: probability of measuring the best network (log scale; bars show the range over 5 seeds). Right: probability that a measurement is a valid four-site network.*

7.3 Comparison with exact and classical methods

The score is a prediction. To test it independently we simulated locating the later aftershocks: arrival times were computed with a different wave speed (5.8 km/s instead of the planner's 6.0) and random timing noise was added, then each aftershock was located by least squares with the permanent stations plus the four chosen sites. We did this for every one of the 210 networks.

Table 15. *Simulated location error on 217 later aftershocks.*

Method	Sites	Rank by score	Mean error (km)	90th percentile (km)
No new sensors	–	–	3.00	–
Average of all 210 networks	–	–	2.60	–
Exact (brute force)	1, 3, 4, 7	1	2.31	3.82
Hybrid QAOA (XY mixer)	1, 3, 4, 7	1	2.31	3.82
Greedy forward	1, 3, 4, 7	1	2.31	3.82
Backward elimination	1, 3, 4, 7	1	2.31	3.82
Simulated annealing	1, 3, 4, 7	1	2.31	3.82
Expert rule (azimuthal gap)	0, 1, 2, 6	159	2.63	4.74
Best by D_N criterion	0, 1, 2, 4	45	2.36	3.98
Random pick (seed 0)	2, 4, 5, 7	44	2.70	4.84

The chosen network lowers the mean location error from 3.00 km to 2.31 km, against 2.63 km for a field rule that closes the largest azimuthal gap. It ranks 4 of 210 in the simulated test (best: 2.30 km); score and test agree reasonably (rank correlation 0.81). Greedy selection, backward elimination and simulated annealing (5 of 5 seeds) find the same answer: this instance is easy for classical methods.

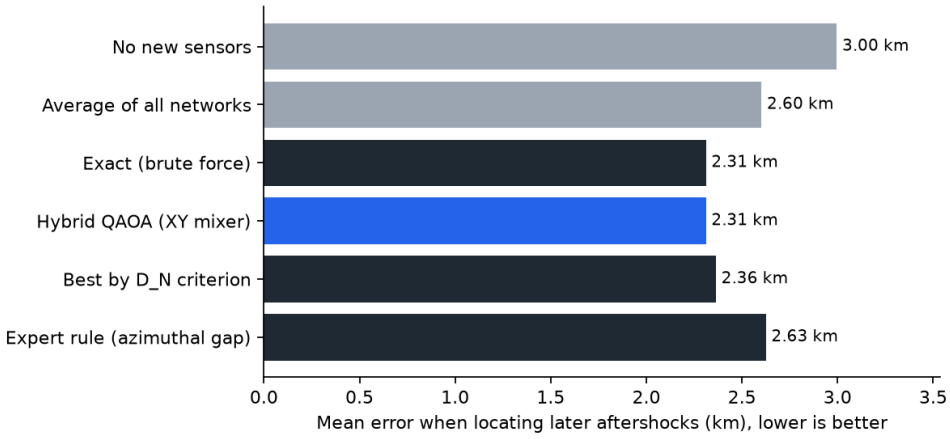


Figure 6. *Mean error when locating later aftershocks in the simulated test (lower is better).*

7.4 The trade-off between accuracy and crew risk

Ignoring risk (λ = 0), the best network is {0, 1, 4, 7} with a total access risk of 2.79. At λ = 150 the optimiser swaps high-risk site 0 for low-risk site 3: risk falls to 2.06 for a small loss in accuracy. Larger λ pushes sensors away from the rupture and accuracy degrades quickly.

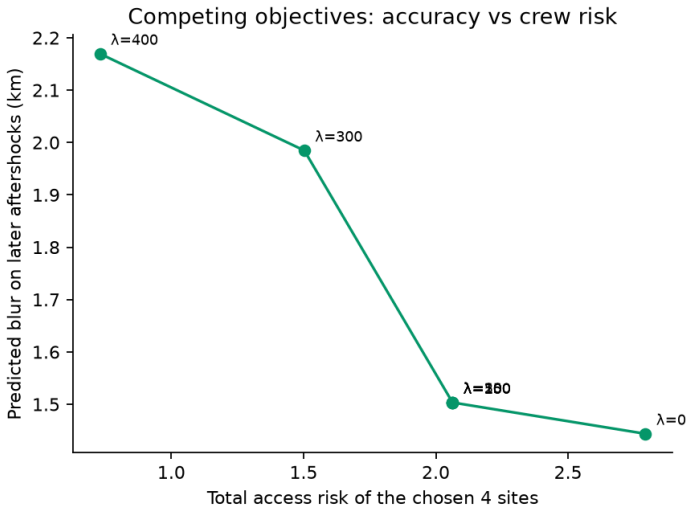


Figure 7. *The exact optimum for different risk weights λ.*

8. Beyond one earthquake

8.1 Five more earthquakes

To check that the method is not tuned to Ridgecrest, we repeated the test on five more earthquakes with 12 hours of real aftershocks, road-checked candidate sites, 3 circuit layers and 3 seeds per variant (11 minutes in total). Ridgecrest is repeated here with a wider station set, so its numbers differ from Chapter 7.

Table 16. *Benchmarks (run_benchmarks.py). Positive 'vs expert' means our optimum located aftershocks better.*

Earthquake	Test events	P(best) mean	P(best) CVaR	P(best) CVaR warm	Exact (CVaR)	vs expert	vs expert, hearing
Ridgecrest 2019	744	3.8%	9.6%	9.6%	3/3	+2.4%	+2.7%
Monte Cristo Range 2020	410	1.2%	7.3%	3.1%	3/3	-0.1%	-0.3%

Earthquake	Test events	P(best) mean	P(best) CVaR	P(best) CVaR warm	Exact (CVaR)	vs expert	vs expert, hearing
Anchorage 2018	338	0.4%	1.6%	1.3%	2/3	-0.7%	-0.0%
Ferndale 2022	20	6.0%	7.0%	10.0%	3/3	-2.9%	-1.6%
Puerto Rico 2020	710	6.2%	8.2%	10.2%	3/3	-15.8%	-15.6%

CVaR training made the final pick exact in 14 of 15 runs, against 8 of 15 with the usual mean-energy training, and raised the chance per measurement on every earthquake. The best chance per measurement in any variant was 10.2%; the 20% target was missed in every run.

> **The result that did not go our way**
>
> In the simulated location test the best-scoring network beat the expert rule on only 1 of 5 earthquakes (Ridgecrest 2019). On Puerto Rico 2020 it was 15.8% worse. The quantum step found the optimum of the score it was given; the score itself (the D-criterion) does not always predict location error. Better placement on these earthquakes needs a better score, not a better search. South Napa 2014 was skipped: only 9 later aftershocks (M >= 2.5, 12 h to 7 days) to test on; at least 10 needed.

Hearing range (the right-hand column): a station whose simulated signal-to-noise ratio is under 5 gives no reading, and weak readings are noisier. It uses the Hutton & Boore (1987) attenuation law with an assumed noise level, so it is a simplification.

8.2 Simulated gate noise

Real quantum hardware makes errors. Without hardware access we simulated depolarising gate errors on a smaller version (6 sites, choose 2, 1 layer) with a Monte-Carlo Pauli-error method in qiskit.quantum_info, 200 trials per point.

Table 17. *Noise study (run_extras.py). Random guessing gives 6.7%.*

Gate error	P(best network)	Valid answers
0 (noise-free)	52.2%	100.0%
0.01%	51.4%	99.2%
0.03%	51.4%	99.2%
0.10%	48.7%	94.9%
0.30%	40.7%	84.5%
1.00%	24.0%	61.0%

At 1% gate error the circuit still beats random guessing clearly, but a third of the answers are no longer valid networks. On hardware the classical check would catch these, at the cost of more shots. This is a classical simulation, not a hardware result.

9. Caveats and limitations

The brief says limitations count in a team's favour. These are ours, grouped by what they affect.

Table 18. Caveats.

Area	Caveat	What it means for our claims
Quantum advantage	With 210 networks, brute force takes milliseconds and greedy search finds the optimum	No speedup is claimed. The instance is small so every answer can be verified
Success rate	The chance per measurement was 5%–10%, below our 20% target	QAOA finds the optimum by sampling many times, not with certainty
Scaling	The exact score needs 385 terms instead of 55, and the count grows fast with the budget	The exact encoding does not scale to large sensor budgets
Circuit depth	About 2,336 at one layer and 6,264 at three, against 54–118 for the penalty circuit	Noise-free simulation only; on current hardware this depth is a serious problem (Section 8.2)
The score	The D-criterion beat the expert rule on only 1 of 5 earthquakes in the location test	The agent's value today is speed, automation and re-planning, not a better score than a field rule
Location test	Simulated: the same simple physics with a different wave speed and added noise, not real recordings	Kilometre figures are relative indicators, not operational errors
Physics	Uniform wave speed, P waves only, fixed picking error, no site effects or 3D structure	Which sites are better is more reliable than the exact numbers
Hearing range	Hutton & Boore attenuation with one assumed noise level for every site	A simplification, opt-in
Inputs	Candidate sites, access risk and λ follow rules we wrote	A response team would set these from local knowledge
Training shortcut	During CVaR training the circuit's output is computed by our exact simulator of the same circuit (tested against Qiskit's Statevector); final measurements run on Qiskit's sampler	Same results, seconds instead of minutes per plan
Live mode	Tested with a recorded feed only; a run cut off by a restart is not resumed	Not yet proven during a real qualifying earthquake
Evidence	'Siting is mostly manual' rests on published reports, not interviews; 'no earlier quantum formulation' means only that our searches found none	We say 'first we found', never 'first ever'

9.1 What did not work or was not built

- The 20% per-measurement success target was missed in every run, on every earthquake.
- The optimised network beat the expert rule on only 1 of 5 earthquakes.
- The penalty-based QAOA did worse than random guessing among valid networks (with one untuned weight).
- Not built from the design: team overrides of individual sites, the nested Monte Carlo check of top designs, a full leave-one-earthquake-out replay, and a run on IBM hardware.
- The three wave-speed weights in memory stay equal in practice: without measured arrival times the models cannot be told apart.
- South Napa 2014 could not be tested: only 9 later aftershocks (M >= 2.5, 12 h to 7 days) to test on; at least 10 needed.

9.2 What we would do next

- A score that predicts location error better than the D-criterion, chosen by held-out error across earthquakes.
- A layered velocity model and real access data (roads, land permits).
- Larger instances (20 to 30 sites) where exhaustive search stops being trivial.
- A run on real IBM hardware, with error mitigation, to compare against the noise study.

10. How the submission meets the brief

Table 19. *Checklist against the hackathon brief, dos and don'ts.*

Brief item	Where it is met	Status
Identify the problem and who it serves	Chapter 1	Done
Formulate: variables, objective, constraints, a small test instance	Chapter 3	Done
Justify the quantum algorithm	Chapter 4	Done
Implement it in Qiskit	src/qaoa_solver.py (Qiskit 2.6.2)	Done
Evaluate against a classical or exact baseline	Sections 7.2, 7.3 and 8.1	Done
Reflect on limitations	Chapter 9	Done
Architecture diagram in the report	Figures 2 and 3	Done
Keep the instance small and simulator-friendly	10 qubits, 210 networks, statevector simulation	Done
Make the quantum part do real work	The circuit chooses every plan; classical code only verifies	Done
README with setup steps	README.md, section 'Run it'	Done
Be honest about what did not work	Sections 8.1 and 9.1	Done
Stay within the six domains	Disaster Response	Done
No copied code; credit reused ideas	Own implementation; ideas credited in References; GPL code deliberately not reused	Done
Video of 3 minutes or less	VIDEO_SCRIPT.md, about 2:50	Team to record
Check every link opens; repository visible to judges	Repository is private until submission	Team to confirm
No commits after the deadline	Last commit before the cutoff	Team to confirm

11. Reproducing the results

All results run offline on a laptop from cached data (Windows 11, Python 3.12, Qiskit 2.6.2). Setup steps are in README.md.

Table 20. *Commands.*

Command	What it produces	Time
python -m pytest tests -q	303 automated checks	about 45 s
python run_experiment.py	Ridgecrest experiment: results/results.json and figures	43 min
python run_benchmarks.py	Five more earthquakes: results/benchmarks.json	11 min
python run_extras.py	Noise study and agent replay: results/extras.json	44 s
python report/make_diagram.py	Figures 2 and 3	seconds
python report/build_report.py	This report	seconds

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Credits and use of AI tools

All code was written for this hackathon; no code was copied from tutorials or other teams. Published ideas (the Dicke-state circuit, the XY mixer, the D-criterion, the attenuation law and the San Martín et al. pattern) are credited above.

The code and documents were written with AI assistance (Claude Code), directed and reviewed by the team. The application itself contains no AI or LLM features.

Appendix A. Glossary

Table 21. *Terms used in this report.*

Term	Meaning
Aftershock	A smaller earthquake that follows a large one in the same area
Azimuthal gap	The largest angle, seen from an earthquake, with no seismometer; a common field rule closes it
CVaR	Conditional value at risk: here, the average energy of the best 10% of measured networks
D-criterion	The log-determinant of the information matrix; larger means a smaller location error ellipsoid
Dicke state	A quantum state that is an equal mix of all bit strings with exactly k ones
Hamiltonian	The energy function the circuit minimises; its lowest state is the best network
Hearing range	Whether a station can detect an aftershock, from its simulated signal-to-noise ratio
QAOA	Quantum Approximate Optimization Algorithm: alternating cost and mixer layers tuned by a classical optimiser
QUBO	Quadratic unconstrained binary optimisation: a score over yes/no variables with pairwise terms
Qubit	A quantum bit; here, one qubit per candidate site
Statevector simulation	An exact classical simulation of a quantum circuit, without noise
XY mixer	A two-qubit gate that swaps a 1 and a 0, so the number of chosen sites stays fixed